1	**Research Topic**	A Comparative Evaluation of Context Engineering Approaches for the Modernization of Legacy Enterprise Web Applications.
2	**Identified Research Gap and Justification**	The modernization of legacy software applications is an ongoing and expensive problem faced by enterprises. AI presents an opportunity to make this process more efficient and affordable. However, improvements in the accuracy of the methods used to translate legacy code to modern standards are required before enterprises fully trust AI driven migration processes to replatform critical applications. A study is required to compare the performance of different AI assisted migration approaches which measures the accuracy of each approach.
3	**Research Problem**	How can context engineering be applied to the modernization of legacy enterprise web applications to achieve new system implementations that are of high software engineering quality, functionally equivalent and continue to reliably meet the requirements of the enterprise?
4	**Research Goals**	1. Identify a reliable context engineering approach for the migration of legacy enterpise web applications to modernized implementations which can be adopted by software engineering teams. 2. Determine if using a structured context engineering approach such as multi-step orchestration workflows or specification driven workflows improves resulting code quality versus using an agent platform with prompt engineering techniques only. 3. Evaluate the best performing combination of agent platform and context engineering workflow. 4. Compare results to similar studies of AI assisted code migration in the literature.
5	**Research Questions**	1. How does the choice of context engineering workflow, or combination of workflows affect resulting migrated code quality and functional accuracy? 2. How does the choice of agent platform affect resulting migrated code quality and functional accuracy? 3. How do the results of the AI assisted code migration experiments compare with previous studies and has this study improved on previous results?

6	**Research Methods for Achieving This Goal**	1. Migrate benchmark legacy java web application benchmark repository code to a modern kotlin and react implementation with the agent platform and context engineering workflow as independent variables.The context engineering workflow independent variable options will be Prompt Engineering, Agent Skills, RAG, Muli-step orchestration workflow (BMAD), Specification driven workflows (SpecKit and OpenSpec). The agent platform independent variables will be Claude Code, GitHub Co-Pilot and Cursor. All other variables including the choice of LLM will be kept constant as dependent variables. Note: The proposed context engineering workflows are progressions with some later approaches such as BMAD, SpecKit and OpenSpec also using RAG or Agent Skills. 2. Evaluate the resulting code quality with appropriate quantitative and qualitative metrics to identify the best performing combination of independent variables. 3. Compare results with similar benchmark studies.
7	**Type of Data Collected, Data Analysis Methods, and Triangulation Potential**	**Data:** The type of data will be a benchmark repository containing legacy java web application code with both frontend, backend and database components. **Analysis Methods:** The analysis method will assess the resulting modernized code output using quantitative and qualitative metrics. **Triangulation:** Mulitple quantitative and qualitative assessment metrics will be used to achieve triangulation and strengthen the conclusions of the study.
8	**Possible Expected Results and Hypotheses**	Context engineering approaches that use AI skills or RAG will achieve better quantitative and qualitative results than prompt engineering. Progressively more structured context engineering approaches such as Multi-Step orchestrated workflows (BMAD) or specification driven approaches (SpecKit, OpenSpec) will achieve better quantitative and qualitative results than prompt engineering, AI skills and RAG. It is expected that one agent platform will achieve better quantitative and qualitative results. The study may yield better results than previous studies migrating the benchmark legacy code repositories which did not apply context engineering workflows.